In certain cases, the immune system gives a negative reaction, in response to the clotting factors when used as a treatment for bleeding. The immune system develops proteins (inhibitors) that inactivate the clotting factors, making treatment less effective.

7.4 ALBINISM

Albinism is a rare group of congenital disorders that cause the skin, hair, or eyes to have little or no colour. Melanin is the pigment responsible for the colour of the skin, hair, and eyes. The term albinism is from the Latin *albus*, means "white". Albinism is associated with a number of vision defects, such as photophobia (abnormal intolerance to visual perception of light), nystagmus(involuntary eye movement, acquired in infancy or later in life, that may result in reduced or limited vision) and amblyopia (disorder of eye sight due to the eye and brain not working well together). Lack of skin pigmentation makes the individual for more susceptibile to sunburn and skin cancers.

There are different types of albinism. Defects in different gene characterize its types.

OCA1 : Defect in tyrosinase and possesses two subtypes , OCA1a and OCA1b.

OCA1a: Subjects with OCA1a have a total absence of the pigment melanin. Therefore the affected individuals have white hair, very pale skin, and light coloured eyes.

OCA1b: Subjects with OCA1b make some melanin. They have light-colored skin, hair, and eyes.

OCA2: This type is common in African descendants and Native American populations. It is less severe than OCA1 and there is reduced melanin production. Individuals with OCA2 are born with light colored skin. Their hair may be yellow, blonde, or light brown.

OCA3: A defect in the TYRP1 gene. This type affects individual with dark skin especially, South African populations. It usually affects people with dark skin, particularly black South Africans. These individuals have reddish-brown skin, reddish hair, and brown eyes.

OCA4: It is due to a defect in the SLC45A2 protein. Like other types, it results in impeded melanin production and common in East Asian descendants.

7.4.1 Causes

A defect in one of several genes described above that produces melanin causes albinism. Melanin are natural pigments found in humans and animals. In the human eye, melanin is present in the uveal tract and the pigmented epithelial layer of the retina. It protects eyes by absorbing visible light that penetrate the lens through binding free radicals. The defective gene passes down from both parents to the child and leads to albininism.

The synthesis of melanin from tyrosine is as shown in the figure 7.7. The figure depicts that melanin is synthesized from tyrosine through DOPA. DOPA is the product of the enzyme, tyrosinase (diphenol oxidase). It is a copper containing enzyme and uses

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